

Huawei TechArena 2025: Challenge Summary and Technical Plan

Coding Assistant Brief

October 31, 2025

1 Challenge Context

Primary objective: Develop a BESS to optimize financial performance by participating in day-ahead, FCR and aFRR markets.

The challenge is divided into three core, interconnected optimization tasks:

1. **Operation Optimization:** Maximize the BESS's revenue over the year 2024 by developing an optimal charge/discharge strategy to bid in the DA, FCR and aFRR markets.
2. **Investment Optimization:** Identify which of five European countries (Germany, Austria, Switzerland, Hungary, Czech Republic) offers the highest Return on Investment (ROI) for installing the BESS over a 10-year period.
3. **Configuration Optimization:** Determine the optimal BESS configuration by analyzing the impact of different C-rates and daily cycle limits on profitability and performance.

Table 1: European Market Participation Rules Summary

Feature	Day-Ahead Market (EPEX SPOT)	FCR Capacity Market	aFRR Capacity Market
Mechanism	Blind Auction	Daily Auction	Daily Auction
Gate Closure Time (D-1)	12:00 CET	8:00 CET	9:00 CET
Product Granularity	15 minutes	4 hours	4 hours
Bid Structure	Energy (MWh)	Symmetric Capacity (MW)	Asymmetric Capacity (MW)
Remuneration	Pay-as-Cleared (Energy)	Pay-as-Cleared (Capacity)	Pay-as-Bid (Capacity)
Minimum Bid Size	0.1 MW	1 MW	1 MW

2 Evaluation Criteria

Submissions will be evaluated based on the following criteria:

1. Revenue maximization (50%). This will evaluate how well the algorithm maximizes revenue based on market prices. A linear scaling formula will be used to map all submission values between the minimum and maximum observed
2. Investment optimization (20%). This will evaluate the assessment of optimal investment locations and markets. The evaluation will consider
3. Configuration optimization (20%). This will evaluate the assessment of the most relevant configuration parameters on BESS revenue.
4. Code Quality and Documentation (10%). This will evaluate the clarity and structure of the code.

Table 2: BESS Technical and Configuration Parameters

Parameter	Symbol	Value	Unit	Source
Fixed Parameters				
Nominal Energy Capacity	E_{nom}	4472	kWh	
Rated Power	P_{rated}	2236	kW	
Initial Investment Cost	C_{inv}	200	EUR/kWh	
Configuration Scenarios				
<i>C-rate = 0.25 C</i>				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.25C}$	$0.25 \times 4472 = 1118$	kW	
<i>C-rate = 0.33 C</i>				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.33C}$	$0.33 \times 4472 \approx 1475.8$	kW	
<i>C-rate = 0.50 C</i>				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.50C}$	$0.50 \times 4472 = 2236$	kW	
<i>Cycles = 1.0/day</i>				
Max Daily Discharge	$E_{\text{dis,max}}^{1.0}$	$1.0 \times 4472 = 4472$	kWh	
<i>Cycles = 1.5/day</i>				
Max Daily Discharge	$E_{\text{dis,max}}^{1.5}$	$1.5 \times 4472 = 6708$	kWh	
<i>Cycles = 2.0/day</i>				
Max Daily Discharge	$E_{\text{dis,max}}^{2.0}$	$2.0 \times 4472 = 8944$	kWh	

3 Project Plan

Outlook

The Interdependence of the Three Optimization Tasks A deeper analysis reveals a nested, hierarchical relationship between them that must be reflected in the project’s modeling strategy.

The ”Optimal Configuration” task is at the base of the hierarchy. The choice of a C-rate (0.5C, 0.33C, 0.25C) and a daily cycle limit (1.0, 1.5, 2.0) directly defines the physical constraints of the BESS, such as its maximum charge/discharge power and daily energy throughput. These parameters are fundamental inputs for the ”Optimal Operation” model.

The ”**Optimal Operation**” model forms the core of the analysis. It takes the configuration parameters as given and, based on market price data, calculates the maximum possible annual revenue for that specific BESS configuration in a given country. This revenue figure is the most critical output of the entire project.

Finally, the ”Optimal Investment” task sits at the top of the hierarchy. It uses the annual revenue generated by the operational model as its primary input to calculate the 10-year ROI. The optimal country for investment is simply the one that yields the highest ROI.

The Nested Hierarchy of the Three Optimization Tasks

1. **”Optimal Operation”**: Focus on building a robust and accurate operational optimization model. This model will then be placed within a larger computational framework that systematically iterates through every possible BESS configuration and every target country.
2. **”Optimal Configuration”**: By running the operational model for each combination (e.g., Germany with 0.25C and 1.0 cycle, Germany with 0.5C and 1.5 cycles, Austria with 0.25C and 1.0 cycle, etc.), the model can then populate a complete results matrix.
3. **”Optimal Investment”**: Use the annual revenue generated by the operational model as its primary input to calculate the 10-year ROI. The optimal country for investment is simply the one that yields the highest ROI considering WACC and inflation rates.

Mathematical Model

Objective Function

The objective is to maximize the total net profit over the one-year horizon. This is the sum of day-ahead energy arbitrage revenue and ancillary service capacity payments, minus the cost of energy purchased for charging.

$$\begin{aligned} \max Z = & \sum_{t \in T} \left(\frac{P_{DA}(t)}{1000} p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000} p_{\text{ch}}(t) \right) \Delta t \\ & + \sum_{b \in B} \left(P_{FCR}(b) c_{fcr}(b) + P_{aFRR}^{\text{pos}}(b) c_{aFRR}^{\text{pos}}(b) + P_{aFRR}^{\text{neg}}(b) c_{aFRR}^{\text{neg}}(b) \right) \Delta b \end{aligned} \quad (1)$$

- The first term is day-ahead net profit. $p_{\text{dis}}(t)$ and $p_{\text{ch}}(t)$ are in kW, $P_{DA}(t)$ is in EUR/MWh; division by 1000 converts kW to MW. Multiplying by Δt (0.25 h) yields EUR per interval.
- The second term is ancillary service capacity revenue. Bids $c(b)$ are in MW and prices $P(b)$ are in EUR/MW for each 4-hour block; multiplying by Δb (4 h) yields EUR per block.

Constraint Equations with Detailed Explanations

The optimization is subject to the following constraints.

1. **State of Charge (SOC) Dynamics:** Energy balance of the BESS.

$$e_{\text{soc}}(t) = e_{\text{soc}}(t-1) + \left(p_{\text{ch}}(t) \eta_{\text{ch}} - \frac{p_{\text{dis}}(t)}{\eta_{\text{dis}}} \right) \Delta t \quad \forall t \in T \quad (2)$$

For $t = 1$, use the initial SOC:

$$e_{\text{soc}}(1) = e_{\text{soc}}^{\text{init}} + \left(p_{\text{ch}}(1) \eta_{\text{ch}} - \frac{p_{\text{dis}}(1)}{\eta_{\text{dis}}} \right) \Delta t. \quad (3)$$

2. **SOC Limits:** Stay within operational boundaries.

$$SOC_{\min} E_{\text{nom}} \leq e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} \quad \forall t \in T \quad (4)$$

3. **Power Limits:** Binary-continuous linking to configured power.

$$p_{\text{ch}}(t) \leq y_{\text{ch}}(t) P_{\max}^{\text{config}} \quad \forall t \in T \quad (5)$$

$$p_{\text{dis}}(t) \leq y_{\text{dis}}(t) P_{\max}^{\text{config}} \quad \forall t \in T \quad (6)$$

4. **Simultaneous Operation Prevention:** No charge and discharge at the same time.

$$y_{\text{ch}}(t) + y_{\text{dis}}(t) \leq 1 \quad \forall t \in T \quad (7)$$

5. **Market Co-optimization Power Limits:** Allocate total power between energy and reserves (convert MW bids to kW by $\times 1000$).

$$p_{\text{ch}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b) \leq P_{\max}^{\text{config}} \quad \forall b \in B, \forall t \in T \quad (8)$$

$$p_{\text{dis}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b) \leq P_{\max}^{\text{config}} \quad \forall b \in B, \forall t \in T \quad (9)$$

6. **Daily Cycle Limit:** Limit the total daily discharged energy.

$$\sum_{t \in d} p_{\text{dis}}(t) \Delta t \leq N_{\text{cycles}} E_{\text{nom}} \quad \forall d \in D \quad (10)$$

7. **Ancillary Service Energy Reserve:** Maintain sufficient energy to deliver awarded capacity for 15 minutes ($\Delta t = 0.25$ h).

$$e_{\text{soc}}(t) \geq SOC_{\min} E_{\text{nom}} + (1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b)) \Delta t \quad \forall b \in B, \forall t \in b \quad (11)$$

$$e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} - (1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b)) \Delta t \quad \forall b \in B, \forall t \in b \quad (12)$$

8. **Minimum Bid Size Constraints:** Bids can be non-zero only if the corresponding binary is 1; if non-zero, they must satisfy minimum size and available power.

$$c_{fcr}(b) \geq y_{fcr}(b) \text{MinBid}_{fcr} \quad \forall b \in B \quad (13)$$

$$c_{fcr}(b) \leq y_{fcr}(b) \frac{P_{\max}^{\text{config}}}{1000} \quad \forall b \in B \quad (14)$$

$$c_{aFRR}^{\text{pos}}(b) \geq y_{aFRR}^{\text{pos}}(b) \text{MinBid}_{aFRR} \quad \forall b \in B \quad (15)$$

$$c_{aFRR}^{\text{pos}}(b) \leq y_{aFRR}^{\text{pos}}(b) \frac{P_{\max}^{\text{config}}}{1000} \quad \forall b \in B \quad (16)$$

$$c_{aFRR}^{\text{neg}}(b) \geq y_{aFRR}^{\text{neg}}(b) \text{MinBid}_{aFRR} \quad \forall b \in B \quad (17)$$

$$c_{aFRR}^{\text{neg}}(b) \leq y_{aFRR}^{\text{neg}}(b) \frac{P_{\max}^{\text{config}}}{1000} \quad \forall b \in B \quad (18)$$

9. **Ancillary Service Market Mutual Exclusivity:** Prevent simultaneous bidding in multiple ancillary service markets for the same block.

$$y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b \in B \quad (19)$$

4 Investment Optimization

This section outlines the investment optimization framework used to evaluate the financial viability of deploying a Battery Energy Storage

4.1 Objective

The primary objective of the investment optimization is to evaluate and rank the financial viability of deploying a Battery Energy Storage System (BESS) across various European countries and technical configurations. This is achieved by performing a 10-year Discounted Cash Flow (DCF) analysis to determine the Net Present Value (NPV) and the Levelized Return on Investment (ROI) for each scenario.

4.2 Financial Modeling Approach

As established, the core of our analysis is a DCF model that correctly aligns nominal cash flows with a nominal discount rate. This approach ensures that the effects of inflation are accounted for consistently and accurately.

- **Cash Flows:** We project future profits by growing the initial year's profit (from the operational optimization) at the country-specific inflation rate. This creates a stream of **nominal cash flows**.
- **Discount Rate:** We use the country-specific Weighted Average Cost of Capital (WACC) as the **nominal discount rate**.

4.3 Step-by-Step Calculation

The analysis follows a three-step process for the results from 45 scenarios (5 countries $\times$ 9 configurations).

4.3.1 Step 1: Projecting Nominal Profits

First, for each country among the five, we take the BEST annual profit of this country (that is, the highest annual profit of nine different configuration settings) calculated by the Pyomo model for the year 2024, denoted as Π_{2024} , and project it over a 10-year horizon from year $y = 1$ (2024) to $y = 10$ (2033). The nominal profit for any future year y is calculated by applying the country's annual inflation rate, π .

$$\Pi_y = \Pi_{2024} \cdot (1 + \pi)^{y-1} \quad (20)$$

4.3.2 Step 2: Calculating Net Present Value (NPV)

Next, we calculate the NPV by summing the discounted values of all future nominal profits and subtracting the initial Capital Expenditure (CAPEX). The CAPEX is given as \$200 per kWh. For a 1 MW BESS with a nominal energy capacity E_{nom} (in MWh), the CAPEX is $200 \times E_{nom} \times 1000$.

The WACC is denoted by i .

$$\text{NPV} = \left(\sum_{y=1}^{10} \frac{\Pi_y}{(1+i)^y} \right) - \text{CAPEX} \quad (21)$$

4.3.3 Step 3: Calculating Levelized Return on Investment (ROI)

Finally, we calculate the Levelized ROI as specified in the competition guidelines. This metric annualizes the return relative to the initial investment.

$$\text{Levelized ROI}(\%) = \frac{\text{PV}(\text{Total Profits})}{\text{CAPEX} \times \text{Lifetime}} \times 100 \quad (22)$$

Where:

- PV(Total Profits) is the first term in the NPV equation: $\sum_{y=1}^{10} \frac{\Pi_y}{(1+i)^y}$.
- CAPEX is the initial investment cost.
- Lifetime is the project duration, which is 10 years.

4.4 Example Calculation

Let's consider a hypothetical scenario for a 1 MW BESS:

- **BESS Configuration:** C-rate = 0.5 C $\implies E_{nom} = 2$ MWh
- **Initial CAPEX:** 200 EUR/kWh $\times$ 2000 kWh = 400,000 EUR
- **Country Parameters:** WACC (i) = 7.0%, Inflation (π) = 2.0%
- **Simulated Profit (2024):** $\Pi_{2024} = 50,000$ EUR

The Levelized ROI is then:

$$\text{Levelized ROI} = \frac{395,437.12}{400,000 \times 10} \times 100 = \frac{395,437.12}{4,000,000} \times 100 = 9.89\%$$

This result indicates that, for this specific scenario, the project yields an average annual return of 9.89% on its initial investment in present value terms, but it does not break even ($\text{NPV} < 0$).

5 Advanced Battery Degradation Modeling for Optimal Dispatch

The economic viability and operational efficacy of a Battery Energy Storage System (BESS) are inextricably linked to the management of its degradation. As electrochemical assets, batteries have a finite operational life that is consumed through usage. An unsophisticated operational model that neglects this consumption of life will invariably lead to premature failure and a negative return on investment.

Conversely, an overly conservative model will fail to capture profitable market opportunities, similarly undermining the asset’s economic case. The central challenge, therefore, lies in developing a degradation model that is both physically accurate and computationally tractable, allowing for its direct integration into an economic dispatch optimization. This section details a progression of modeling techniques, from foundational cost-based approximations to high-fidelity, direct life-loss formulations, culminating in a proposed unified sub-model for the Phase II architecture.

5.1 The Foundational Approach: Piecewise Linear Cost of Cycle Depth

The initial step in translating physical battery wear into an economic signal involves creating a cost function that can be incorporated into a standard market dispatch program. The non-linear relationship between the depth of a charge-discharge cycle and the resultant aging is a well-documented phenomenon in lithium-ion batteries.[1] Deeper cycles inflict disproportionately more damage than shallower ones. For instance, a cell might endure 50,000 cycles at a 10% depth of cycle (DOC), but only 500 cycles at a 100% DOC.[1] This non-linearity poses a challenge for linear and mixed-integer linear programming (MILP) models, which are the standard for market clearing and economic dispatch.

A seminal approach to address this challenge, proposed by Xu et al., is to approximate this non-linear aging characteristic with a piecewise linear cost function.[1] This method provides a tractable yet reasonably accurate representation of cycle aging that can be “incorporated easily into existing market dispatch programs”.[1] The core idea is to divide the full range of possible cycle depths (0% to 100%) into a series of discrete segments. Within each segment, the marginal cost of aging is assumed to be constant. The resulting cost curve is a convex, stepwise function that approximates the true, smooth, non-linear aging curve.

The mathematical formulation begins with the cycle depth stress function, $\Phi(\delta)$, which describes the percentage of life lost for a single cycle of depth δ . For the Li(NiMnCo)O₂ cells studied, this function is near-quadratic, e.g., $\Phi(\delta) = (5.24 \times 10^{-4})\delta^{2.03}$.[1] To convert this physical life loss into a cost, the model prorates the total battery cell replacement cost, R (in dollars), over the total expected life. The marginal aging cost, c_j , for each segment j of the piecewise linear function is then calculated as:

$$c_j = \frac{R}{\eta^{dis} E^{rate}} J [\Phi(\frac{j}{J}) - \Phi(\frac{j-1}{J})] \quad (23)$$

where E^{rate} is the energy capacity of the BESS, η^{dis} is the discharge efficiency, and J is the total number of linear segments.[1] This equation calculates the incremental life loss associated with moving from the start to the end of a segment, converts it to a cost, and expresses it as a marginal cost per unit of energy discharged within that segment.

The effectiveness of this approximation is directly proportional to the number of segments, J . A single-segment model provides a very coarse, constant marginal cost, leading to overly conservative dispatch. As the number of segments increases (e.g., to 4, 8, or 16), the piecewise linear function more closely tracks the true non-linear curve, allowing the dispatch model to make more nuanced trade-offs between arbitrage revenue and aging cost. With a sufficient number of segments, the modeling error compared to a benchmark non-linear model approaches zero.[1] This approach, while foundational, serves as the conceptual basis for integrating more complex degradation physics into an economic framework.

5.2 A Multi-Factor Approach: Linearizing Calendar and Cyclic Aging Stressors

While cycle depth is a primary driver of battery degradation, it is not the only one. A more comprehensive model must account for the distinct physical mechanisms of calendar and cyclic aging, each influenced by a different set of operational stress factors. The work by Collath et al. provides a significant advancement by developing linearized models for both phenomena, allowing a multi-factor degradation assessment to be embedded within an MILP optimization framework.[1]

Calendar Aging: This form of degradation occurs continuously, even when the BESS is idle, and is primarily driven by the battery’s State of Charge (SOC) and temperature. Storing a battery at a high SOC accelerates parasitic chemical reactions, such as the growth of the solid electrolyte interphase (SEI), leading to irreversible capacity loss.[1] To capture this, a linearized model of calendar aging is formulated. The non-linear relationship between SOC and the rate of calendar capacity loss is approximated using a piecewise linear function, which is implemented in the MILP using Special Ordered Sets of Type 2 (SOS2) variables. The formulation is structured as follows:

$$\sum_{i \in I} \lambda_{t,i}^{cal} \cdot X_i^{cal} = soc_t \quad \forall t \in T \quad (24)$$

$$\sum_{i \in I} \lambda_{t,i}^{cal} \cdot Z_i^{cal} = q_t^{loss,cal} \quad \forall t \in T \quad (25)$$

$$\sum_{i \in I} \lambda_{t,i}^{cal} = 1 \quad \forall t \in T \quad (26)$$

Here, for each time step t , soc_t is the state of charge. The variables $\lambda_{t,i}^{cal}$ are the SOS2 weights, where at most two adjacent weights can be non-zero. X_i^{cal} and Z_i^{cal} are the constant coordinates (SOC and corresponding capacity loss rate, respectively) of the breakpoints in the piecewise linear function. The variable $q_t^{loss,cal}$ represents the resulting calendar capacity loss in that time step.[1] This formulation incentivizes the BESS to minimize time spent at high SOC levels, a behavior that a simple cycle-depth model would not capture.

Cyclic Aging: This form of degradation is a direct result of charging and discharging the battery. The model from Collath et al. expands upon the cycle-depth-only approach by also considering the charge-discharge rate (C-rate). High C-rates can induce additional stress mechanisms like lithium plating, accelerating degradation.[1] Similar to the calendar aging model, the complex, multi-variable function describing cyclic aging is linearized. The optimization horizon is divided into blocks (e.g., 4-hour blocks), and the total energy throughput in each block is used as a proxy for the combination of DOC and C-rate. This relationship is then linearized using another set of SOS2 variables ($\lambda_{h,j}^{cyc,ch}$ and $\lambda_{h,j}^{cyc,dis}$), allowing the model to calculate the cyclic capacity loss ($q_h^{loss,cyc,ch}$ and $q_h^{loss,cyc,dis}$) for each block h . [1]

By disaggregating degradation into these two components, the optimization model gains a much more nuanced understanding of battery physics. It can now differentiate between two dispatch schedules that have the same total energy throughput but different operational profiles. For example, it will correctly identify that a schedule involving long periods of storage at high SOC is more damaging than one that cycles the same amount of energy but maintains a lower average SOC. This increased fidelity translates directly into improved economic outcomes, with studies showing that incorporating these models can increase lifetime profitability by over 20% compared to a simple energy throughput model.[1]

5.3 Integrating Cycle History: A Linearized Rain-Flow Cycle Counting (RFCC) Formulation

The models presented thus far approximate cyclic aging based on the characteristics of individual or aggregated cycles. However, the true nature of fatigue damage in materials, including battery electrodes, is cumulative and path-dependent. An irregular SOC profile, typical of real-world BESS operation, consists of numerous cycles of varying depths and mean SOC's superimposed on one another. The Rain-Flow Cycle Counting (RFCC) algorithm is widely recognized as the industry-standard benchmark for accurately identifying and quantifying the cumulative damage from such irregular load histories.[1, 1]

The primary obstacle to using RFCC in optimization has been its procedural nature; the algorithm is a process applied to a time-series profile and "cannot be integrated directly within an optimization problem".[1] Consequently, it has historically been used only as an *ex-post* tool to validate the results of a dispatch optimization, not to guide the optimization itself.[1]

A significant methodological breakthrough, presented by Soleimani et al., is the development of a **linearized RFCC method** that can be directly embedded within an MILP formulation.[1] This approach bridges the gap between the benchmark methodology and practical optimization. The process involves several steps:

1. **Obtain the Manufacturer's Data:** Start with the battery's cycle-to-failure versus DoD curve, which provides discrete data points on the battery's lifetime under regular cycling conditions.[1]
2. **Fit a Continuous Function:** To account for cycle depths not explicitly listed in the datasheet, a continuous function, such as a two-term exponential, is fitted to the data points. This creates a smooth curve, $C_f(\text{DoD})$, representing the number of cycles to failure for any given DoD.[1]

$$C_f = Ae^{B(\text{DoD})} + Ce^{D(\text{DoD})} \quad (27)$$

3. **Invert to Life Loss:** The life loss for a single cycle is the reciprocal of the cycles to failure, $1/C_f(\text{DoD})$. This function represents the fraction of total life consumed by one cycle of a given depth.
4. **Piecewise Linearization:** This non-linear life loss function is then approximated using a piecewise linear function. This is the critical step that makes the model MILP-compatible. The formulation calculates the Loss of Health ($LoH_{bat,t,n}$) for a cycle with a given $DoD_{bat,t,n}$ based on which linear segment the DoD falls into [1]:

$$LoH_{bat,t,n} = \left\{ \frac{\frac{1}{Y_2} - \frac{1}{Y_1}}{X_2 - X_1} (\text{DoD}_{bat,t,n} - X_1) + \frac{1}{Y_1} \right\} \quad (28)$$

where (X_1, Y_1) and (X_2, Y_2) are the coordinates of the start and end points of the relevant linear segment on the (DoD, Cycles-to-failure) curve.

By integrating this formulation, the optimization model can directly calculate the LoH for each charge-discharge cycle it considers. This represents a fundamental shift from the previous methods. Instead of using an abstract marginal cost per MWh discharged, the model now operates with a direct, physically meaningful measure of life consumption. It can accurately assess the damage from the complex, irregular cycling patterns that arise from market arbitrage, enabling a far more precise and intelligent dispatch that truly minimizes long-term battery wear.

5.4 Proposed Degradation Sub-Model for Phase II: A Unified MILP Formulation

To construct the most robust and accurate degradation sub-model for the Phase II architecture, the strongest elements from the reviewed literature must be synthesized into a single, cohesive formulation. The proposed model will therefore be a hybrid that captures the distinct physics of both calendar and cyclic aging with the highest fidelity available within an MILP framework.

For Calendar Aging: The proposed model will adopt the linearized, SOC-dependent formulation from Collath et al..[1] This is the only method presented that explicitly and tractably models the degradation that occurs while the battery is idle. Neglecting calendar aging would lead to a systematic underestimation of total degradation, particularly in applications where the BESS may spend significant time at high states of charge waiting for price spikes. The use of SOS2 variables provides a well-established and computationally efficient method for handling the non-linear SOC dependency.

For Cyclic Aging: The proposed model will adopt the linearized Rain-Flow Cycle Counting (RFCC) model from Soleimani et al..[1] This method is chosen over the simpler cycle-depth or energy-throughput models because of its superior physical fidelity. By directly approximating the benchmark RFCC algorithm, it correctly accounts for the cumulative and path-dependent nature of fatigue damage from irregular cycling. This allows the model to make more intelligent dispatch decisions that cannot be captured by simpler metrics, ultimately leading to better preservation of the battery's health and a longer effective lifetime.

The total degradation within any given time period will be calculated as the sum of the loss from these two components. The total Loss of Health (LoH) in the objective function will be formulated as:

$$LoH_{total} = LoH_{calendar} + LoH_{cyclic} \quad (29)$$

where $LoH_{calendar}$ is determined by the SOC-dependent model and LoH_{cyclic} is determined by the linearized RFCC model. This unified formulation represents the current state-of-the-art in tractable, optimization-embedded battery degradation modeling. It provides a comprehensive accounting of the major stress factors and aging mechanisms, ensuring that the Phase II operational model makes decisions based on a holistic and physically grounded understanding of its impact on the BESS's long-term health. The progression from abstract cost functions to direct, multi-factor, path-dependent LoH calculation signifies a maturation of BESS modeling, enabling more sophisticated and value-driven operational strategies.

6 The Optimization Architecture: Deterministic Model Predictive Control

An advanced degradation model provides the physical foundation for optimal BESS operation, but it must be situated within a control architecture that can navigate the dynamic nature of real-world electricity markets. A deterministic model that assumes perfect knowledge of future prices and renewable generation is brittle if used in an open-loop fashion. However, when combined with the adaptive, corrective capabilities of Model Predictive Control (MPC), it becomes a robust and computationally efficient approach for real-world operation.

6.1 The Deterministic Approach: Optimizing Against a Single Forecast

In a deterministic optimization framework, the model operates on the basis of a single, best-guess forecast for all uncertain variables, such as market prices, load, and renewable generation, over a future time horizon. This approach simplifies the optimization problem significantly compared to stochastic methods, which must consider multiple potential future scenarios. The core assumption is that this single forecast is the most likely representation of future conditions.

While this simplification makes the problem more computationally tractable—a key consideration when using open-source solvers—it does not ignore the reality of forecast errors. The inevitable deviations between the forecast and actual outcomes are managed by the overarching control strategy, as detailed in the following section.

6.2 Ensuring Operational Viability: The Model Predictive Control (MPC) Framework

The Model Predictive Control (MPC) framework provides the mechanism for real-world execution, adaptation, and correction. The MPC architecture, as proposed by Collath et al., addresses the fact that no matter how sophisticated the forecast, it will inevitably have errors, and the optimization model is still a simplified representation of reality.[1]

The MPC architecture operates as a closed-loop, rolling-horizon control strategy. The process is iterative:

1. **Optimization:** At the current time t , the system solves a deterministic optimization problem over a defined future prediction horizon (e.g., 12 or 24 hours), using the most up-to-date system state and point forecasts.
2. **Implementation:** Instead of implementing the entire 24-hour plan, only the decision for the very next control interval (e.g., the next 5 or 15 minutes) is extracted from the optimal solution and sent as a setpoint to the BESS.
3. **State Update & Measurement:** Time advances to the next interval, $t+1$. The system measures the true state of the BESS (its actual SOC and temperature) and obtains new, updated forecasts for prices, load, and generation for the next 24 hours.
4. **Iteration:** The process repeats from step 1, re-solving the optimization problem with the updated information.

A critical component of this framework is the use of a high-fidelity **“digital twin”** of the BESS.[1] This is a detailed simulation model that accurately represents the battery’s electrical and thermal behavior, as well as its degradation physics. When the optimization model produces a dispatch signal, this signal is first fed to the digital twin. The digital twin simulates the BESS’s response and calculates the resulting end-of-interval SOC and, crucially, the updated State of Health (SOH). This updated state is then used as the initial condition for the next optimization run.

This MPC loop provides two key benefits. First, it makes the system resilient to forecast errors. As forecasts are updated at each step, the plan is continuously corrected. Second, it addresses the problem of model mismatch. The optimization model itself is a simplification (e.g., using linearized efficiencies or degradation models). The digital twin, being more accurate, provides a “ground truth” feedback mechanism, preventing the simplified model’s state from drifting away from the physical reality of the battery over time.[1]

6.3 Proposed Phase II Architecture: A Deterministic MPC Framework

The proposed architecture for the Phase II model is a **Deterministic Model Predictive Control (DMPC)** system. This architecture leverages the computational efficiency of deterministic optimization with the adaptive strengths of the MPC loop to create a control system that is both practical and robust for real-world execution.

The operational flow of the DMPC is as follows: At each time step of the MPC loop (e.g., every 15 minutes), the controller solves a single deterministic MILP as described in the subsequent sections. This optimization uses the current SOH and SOC from the digital twin and the latest point forecasts to determine an optimal 24-hour dispatch plan. From this plan, only the decision for the immediate next time step is implemented. The digital twin then calculates the resulting state, new forecasts are obtained, and the entire process repeats.

This architecture effectively addresses two distinct but critical types of modeling error. The frequent re-optimization of the MPC loop addresses **input uncertainty**—the inherent errors in forecasting future market and system conditions. By constantly updating its plan with new information, the system adapts to reality as it unfolds. The feedback from the high-fidelity digital twin addresses **model uncertainty**—the inevitable discrepancies between the simplified, linearized optimization model and the complex, non-linear reality of the physical BESS.

Furthermore, this DMPC architecture provides a natural framework for long-term adaptive control. As the real-world BESS ages, its physical parameters (e.g., capacity, internal resistance) will change. These changes can be identified through periodic measurements and used to update the parameters of the digital twin. Because the optimization model receives its initial state from this continuously updated digital twin, its dispatch strategies will automatically adapt to the battery’s true, evolving State of Health. This ensures that the BESS operation remains optimal not just at the beginning of its life, but throughout its entire operational lifespan.

7 Lifetime Value Maximization: The Objective Function

The technical sophistication of the degradation and control models is ultimately in service of an economic goal: maximizing the financial return of the BESS asset. A purely technical objective, such as minimizing degradation, is suboptimal as it ignores the revenue-generating purpose of the battery. This section redefines the economic objective, moving beyond simple cost minimization to the strategic maximization of lifetime economic value. This is achieved by treating the cost of aging not as a fixed physical parameter, but as a tunable economic lever that balances short-term profit against long-term asset health.

7.1 Quantifying Degradation: From Physical Loss to Marginal Aging Cost

The first step is to translate the physical degradation calculated by the models in Section 1 into a monetary value that can be included in the objective function. Whether calculated as a fractional Loss of Health (LoH) from the linearized RFCC model [1] or as a capacity loss (q^{loss}) from the multi-factor model [1], this physical quantity must be assigned a cost. This is achieved by introducing a key parameter: the **aging cost** (c^{aging}), typically expressed in dollars per kWh of capacity loss or a similar unit.

The cost of aging term in the objective function is then formulated as a simple product:

$$\text{Cost}_{\text{aging}} = c^{\text{aging}} \cdot \text{LoH}_{\text{total}} \quad (30)$$

This term creates the central economic trade-off that the optimization must solve at every time step. The model is incentivized to maximize market revenues (e.g., from energy arbitrage) while minimizing the sum of charging costs and this newly quantified aging cost. Every potential action—charging, discharging, or idling—is thus evaluated based on both its immediate revenue potential and its long-term cost in terms of consumed battery life. This explicit trade-off is the core mechanism of any aging-aware operational strategy.

7.2 The Optimal Aging Cost (c^{aging}): A Paradigm Shift in BESS Economics

The conventional approach in many BESS studies is to set the value of c^{aging} equal to a fixed, physically-derived quantity, such as the battery’s initial investment cost or its anticipated replacement cost.[1] However, this approach has a fundamental flaw: the profit generated from a market application is independent of the original system cost. A paradigm-shifting concept, articulated by Collath et al., is

that c^{aging} should not be treated as a fixed physical constant but as a **tunable hyperparameter**—an economic penalty factor that dictates the aggressiveness of the BESS dispatch strategy.[1]

The value of this parameter directly controls the balance between present and future profits:

- **A low c^{aging} value** tells the optimizer that battery health is relatively cheap. The BESS will operate aggressively, cycling frequently to capture even small price spreads. This strategy front-loads profits but leads to rapid degradation and a shorter operational life.
- **A high c^{aging} value** tells the optimizer that battery health is very expensive. The BESS will operate conservatively, remaining idle unless a very large, highly profitable arbitrage opportunity arises. This strategy preserves the battery’s health and extends its life, but at the cost of forgoing smaller, more frequent revenue streams. This back-loads profits.

Neither of these extremes is likely to be optimal. The true goal is to find the specific value of c^{aging} that maximizes the **Net Present Value (NPV)** of the total profit stream over the entire planned investment horizon. The NPV is a standard financial metric that correctly accounts for the time value of money, recognizing that a dollar earned today is more valuable than a dollar earned ten years from now.

The optimal value of c^{aging} is therefore not the replacement cost. It is a complex function of external economic and financial factors. For instance:

- **Market Volatility:** In a highly volatile market with frequent, large price swings, the opportunity cost of being conservative is high. This justifies a more aggressive strategy, corresponding to a lower optimal c^{aging} . [1]
- **Interest Rate (Discount Rate):** A higher interest rate means future profits are discounted more heavily. This places a premium on near-term revenues, again justifying a more aggressive strategy and a lower optimal c^{aging} . [1]

This concept decouples the engineering problem (modeling how the battery degrades) from the economic problem (deciding how to operate it for maximum financial return). The engineering model provides the physical ‘LoH’, but the economic decision of how much to penalize that ‘LoH’ is a strategic choice based on market conditions and financial objectives.

7.3 Proposed Objective Function for Phase II: Maximizing Lifetime Net Present Value

The objective function for the Phase II model will be formulated to maximize the profit over the optimization horizon, explicitly incorporating the tunable aging cost. At each step of the DMPC loop, the optimization problem to be solved is:

$$\max \sum_{t=1}^T (\text{MarketRevenue}_t - \text{ChargingCost}_t) - \sum_{t=1}^T \text{Cost}_t^{\text{aging}} \quad (31)$$

The aging cost itself is defined as:

$$\text{Cost}_t^{\text{aging}} = c^{\text{aging}} \cdot (\text{LoH}_t^{\text{calendar}} + \text{LoH}_t^{\text{cyclic}}) \quad (32)$$

Crucially, the value of c^{aging} used in this operational model is not arbitrary. It must be predetermined through an *outer-loop optimization process*. This process, as demonstrated by Collath et al., uses the full DMPC framework and the high-fidelity digital twin. It involves running multiple full-lifetime simulations (e.g., for a 10- or 15-year investment period) with different candidate values of c^{aging} . The total lifetime NPV is calculated for each simulation run, and the value of c^{aging} that yields the highest NPV is identified as the optimal value.[1] This optimal value is then hard-coded into the operational model for real-world deployment.

To further enhance the economic rationality of the model, an advanced concept of **discounting the aging cost** can be incorporated directly within the optimization. The aging cost parameter is made time-dependent, increasing over the years according to the interest rate i :

$$c_m^{\text{aging}'} = c_{\text{base}}^{\text{aging}} \cdot (1 + i)^m \quad (33)$$

where m is the year of operation.[1] This formulation effectively tells the optimizer that degradation in the early years is "cheaper" in present value terms than degradation in later years. This incentivizes the model to generate more profit in the early years of the project, when its contribution to the NPV is greatest, further aligning the short-term dispatch decisions with the long-term goal of maximizing investment value. This transforms the BESS from a simple energy-shifting device into a strategically managed financial asset whose operational "risk profile" can be tuned to match market conditions and investor objectives.

8 High-Fidelity System and Grid Constraint Modeling

A comprehensive BESS operational model must not only capture the internal physics of the battery and the economics of the market but also respect the physical and operational constraints of the electrical grid to which it is connected. Simpler models often treat the grid as an infinite source and sink (a "copper plate"), ignoring the BESS's potential impact on local network conditions.[1, 1] For many applications, particularly for BESS deployed within distribution networks, this is an oversimplification. A high-fidelity model must incorporate grid physics to ensure its dispatch schedules are feasible and to unlock new value streams related to grid support.

8.1 BESS Physical and Market Participation Constraints

The foundation of the constraint set is the representation of the BESS's own physical and operational limits. These constraints, drawn from the methodologies across the reviewed literature, are essential for any realistic BESS model.

- **State of Charge (SOC) Balance:** This is the core energy accounting equation, tracking the energy level in the battery over time.

$$SoC_{bat,t} = SoC_{bat,t-1} + \eta^{ch} P_{bat,t}^{ch} \Delta t - \frac{1}{\eta^{dch}} P_{bat,t}^{dch} \Delta t \quad (34)$$

where P^{ch} and P^{dch} are the charge and discharge powers, and η^{ch} and η^{dch} are the respective efficiencies.[1]

- **Power and Energy Limits:** The BESS is constrained by its maximum charge/discharge power ratings and its minimum/maximum energy storage levels (SOC limits).

$$\begin{aligned} 0 &\leq P_{bat,t}^{ch} \leq \overline{P^{ch}} \\ 0 &\leq P_{bat,t}^{dch} \leq \overline{P^{dch}} \\ \underline{SoC} &\leq SoC_{bat,t} \leq \overline{SoC} \end{aligned}$$

These constraints define the operational envelope of the asset.[1, 1]

- **Simultaneous Charge/Discharge Prevention:** A physical BESS cannot charge and discharge at the same time. This is enforced in an MILP model using a binary variable, v_t , which takes a value of 1 for discharging and 0 for charging.[1]

$$\begin{aligned} g_t &\leq G \cdot v_t \\ d_t &\leq D \cdot (1 - v_t) \end{aligned}$$

where g_t and d_t are discharge and charge power, and G and D are the power ratings.

- **Ancillary Service Constraints:** When participating in ancillary service markets, such as frequency regulation, the BESS must adhere to specific market rules. For example, North American Electric Reliability Corporation (NERC) standards may require that a resource providing regulation reserve has enough stored energy to sustain its committed reserve capacity for at least one hour. These requirements are modeled as additional linear constraints that link the BESS's energy level (SOC) to its power capacity and reserve commitment.[1]

8.2 A Grid-Aware Formulation: The Linearized AC-Optimal Power Flow (AC-OPF) Model

To move beyond the "copper plate" assumption, the model must incorporate the physics of the electrical grid. A full, non-linear AC Optimal Power Flow (AC-OPF) model is the most accurate representation but is computationally intractable for use in a complex MILP. A DC-OPF is linear and fast but highly simplified, ignoring reactive power, voltage magnitudes, and active power losses.[1]

The approach proposed by Soleimani et al. offers a powerful compromise: a **linearized AC-OPF model**. [1] This method retains many of the key features of a full AC-OPF while remaining compatible with a linear optimization framework. It provides a computationally efficient yet physically realistic representation of a distribution network, making it ideal for grid-aware BESS dispatch. Compared to a full AC-OPF, it is significantly faster and guarantees a globally optimal solution (within the bounds of the linearization), whereas non-linear solvers may only find local optima. Compared to a DC-OPF, it is far more accurate, with studies showing an error in the objective function of only ~1% relative to a full AC-OPF, versus ~10% for a DC-OPF.[1]

The key features modeled by the linearized AC-OPF include:

- **Active and Reactive Power Balance:** Kirchhoff's Current Law is enforced at each bus for both active and reactive power, ensuring that generation, load, and power flows are balanced.
- **Voltage Drops:** Ohm's Law is used to model the voltage drop across network lines as a function of the power flowing through them and their impedance ($R_{ij} + jX_{ij}$).
- **Power Losses:** The model explicitly calculates the active power losses (I^2R) on the network lines.
- **Operational Limits:** The formulation includes constraints for bus voltage limits (V_{min} , V_{max}) and line thermal limits (maximum current, I_{max}).

The non-linear relationships in the power flow equations (e.g., $S^2 = P^2 + Q^2$ and $P = VI \cos \theta$) are handled through various mathematical linearization techniques, such as piecewise approximations.[1]

8.3 Integrating Network Constraints into the BESS Dispatch Problem

The coupling of the BESS model (Section 4.1) and the grid model (Section 4.2) is straightforward but powerful. The BESS's charge and discharge power variables are treated as a negative and positive power injection, respectively, in the power balance equations of the linearized AC-OPF. For the bus i where the BESS is located, the active power balance equation is modified to include the BESS term:

$$P_{i,t}^{\text{Grid}} + P_{i,t}^{\text{Renewable}} + P_{bat,t}^{dch} - P_{bat,t}^{ch} - P_{i,t}^{\text{Load}} - \sum_j P_{ij,t}^{\text{Flow}} = 0 \quad (35)$$

This integration fundamentally changes the nature of the optimization problem. The model's decisions are no longer based solely on market prices and battery health. They are now also constrained by the physics of the local grid. The optimizer must find a dispatch schedule that is not only profitable but also ensures that no bus voltages or line currents violate their operational limits.

This grid-aware formulation unlocks the BESS's potential to provide valuable grid services beyond bulk energy arbitrage. By understanding its impact on the network, the BESS can be dispatched to:

- **Alleviate Thermal Congestion:** Reduce power flow on an overloaded line during peak hours.
- **Provide Voltage Support:** Inject or absorb reactive power to maintain voltages within their statutory limits in areas with high renewable penetration.
- **Defer Network Upgrades:** Act as a non-wires alternative to traditional grid reinforcement by managing local peak demand.

The inclusion of the linearized AC-OPF model represents a strategic choice about the BESS's primary business case. For a large, transmission-connected asset focused purely on wholesale market arbitrage, it may be an unnecessary complexity. However, for a distribution-connected asset, the ability to provide these localized grid services, enabled by a grid-aware dispatch model, may be its primary source of value and revenue. The Phase II model should therefore be designed in a modular fashion, allowing the user to activate the AC-OPF constraint set based on the specific application and location of the BESS.

9 Synthesis and Implementation Roadmap for the Phase II Model

The preceding sections have detailed the advanced components required for a state-of-the-art BESS operational model. This final section synthesizes these components into a single, cohesive mathematical formulation and provides a practical roadmap for its implementation, parameterization, and validation. This serves as a blueprint for translating the theoretical framework into a functional and effective operational tool.

9.1 The Complete Mixed-Integer Linear Programming (MILP) Formulation

The proposed Phase II model is a Deterministic Model Predictive Control (DMPC) system. At each time step of the control loop, a deterministic MILP is solved. The generalized structure of this MILP is as follows:

Objective Function: Maximize the Net Present Value (NPV) of profit over the optimization horizon.

$$\max \left(\sum_{t \in T} (\text{MarketRevenue}_t - \text{ChargingCost}_t) - \sum_{t \in T} c_t^{\text{aging}} \cdot (\text{LoH}_t^{\text{calendar}} + \text{LoH}_t^{\text{cyclic}}) \right) \quad (36)$$

- The first term is the net revenue from energy and ancillary service markets based on a point forecast.
- The second term is the total cost of degradation, calculated using the time-discounted optimal aging cost parameter c_t^{aging} and the LoH from the unified degradation model.

Key Decision Variables:

- BESS charge/discharge power schedule: $P_{bat,t}^{ch}, P_{bat,t}^{dch}$.
- Binary variables for operating state (charge/discharge/idle): v_t .
- Ancillary service commitments: q_t .
- Grid power injections/withdrawals: $P_{i,t}^G, Q_{i,t}^G$.
- Network power flows and losses: $P_{ij,t}, Q_{ij,t}$.
- Bus voltage magnitudes and angles (or their linearized equivalents): $u_{i,t}$.

Constraint Sets:

1. BESS Physical Constraints (Section 4.1):

- SOC balance equation.
- Power and energy (SOC) limits.
- Simultaneous charge/discharge prevention constraints.

2. Market Participation Constraints (Section 4.1):

- Ancillary service sustainability and ramping requirements.

3. Unified Degradation Model Constraints (Section 1.4):

- SOS2 constraints for the linearized calendar aging model.
- Piecewise linear constraints for the linearized RFCC cyclic aging model.

4. Linearized AC-OPF Grid Constraints (Section 4.2, Optional):

- Active and reactive power balance equations at each bus, including BESS injections.
- Linearized voltage drop and power flow equations.
- Bus voltage and line thermal limit constraints.

This comprehensive formulation integrates all the advanced features discussed, providing a powerful tool for optimizing BESS operation in complex, dynamic environments.

9.2 Data Requirements, Parameterization, and Computational Considerations

The implementation of this advanced model requires a comprehensive dataset. The necessary inputs can be categorized as follows:

- **BESS Parameters:**
 - Power and energy ratings (P^{max} , E^{rate}).
 - Charge/discharge efficiencies (η^{ch} , η^{dch}).
 - Initial investment or replacement cost (R) for calculating c^{aging} .
 - Initial SOC and SOH.
- **Degradation Parameters:**
 - For calendar aging: Coefficients for the SOC-dependent loss function, typically derived from laboratory testing or detailed manufacturer datasheets.[1]
 - For cyclic aging: The cycle-to-failure vs. DoD curve, also from manufacturer datasheets or empirical testing.[1]
- **Market and Financial Data:**
 - Deterministic (point) forecasts for energy and ancillary service prices.
 - The financial discount rate (interest rate, i) for NPV calculations and time-discounting of c^{aging} .
- **Grid Data (if AC-OPF module is active):**
 - Network topology (bus-branch model).
 - Line impedances (R_{ij} , X_{ij}) and thermal ratings.
 - Deterministic (point) forecasts for bus-level loads and renewable generation.

Computational Burden: The formulation is a large-scale MILP. Its computational complexity and solution time will be sensitive to several factors: the length of the optimization horizon, the number of segments used in the various piecewise linear approximations, and the size of the grid model. By using a deterministic approach, the computational burden is significantly reduced compared to a stochastic model, making it more amenable to solution with open-source MILP solvers within practical time limits.[1] The choice of a high-performance MILP solver (e.g., CPLEX, Gurobi, or open-source alternatives like CBC or SCIP) is also critical.[1]

9.3 A Strategy for Validation and Hyperparameter Tuning

The final step before deployment is the rigorous validation of the model and the tuning of its key hyperparameter, the optimal aging cost c^{aging} . The DMPC framework itself provides the ideal environment for this process.

Validation: The high-fidelity digital twin is the cornerstone of validation. The operational model (the MILP) is a simplification of reality. By running the DMPC loop, where the MILP’s decisions are executed on the more accurate digital twin, one can assess the performance and robustness of the control strategy. Key metrics to track include the deviation between the predicted SOC/SOH from the MILP and the simulated SOC/SOH from the digital twin, as well as the overall economic performance.

Hyperparameter Tuning (c^{aging}): As established in Section 3, c^{aging} is not a physical constant but a strategic economic parameter. Finding its optimal value is essential for maximizing the BESS’s lifetime value. The following data-driven, simulation-based procedure should be employed:

1. **Develop the Digital Twin:** Construct and validate a high-fidelity digital twin that accurately represents the specific BESS being deployed.
2. **Select Candidate Values:** Define a range of candidate values for the base aging cost, c_{base}^{aging} . This range should span from very conservative (high cost) to very aggressive (low cost).

3. **Run Lifetime Simulations:** For each candidate value of c_{base}^{aging} , execute a full-lifetime simulation (e.g., 15 years) using the complete DMPC framework. This involves running the rolling-horizon optimization loop for the entire simulated period.
4. **Calculate Lifetime NPV:** For each completed simulation, calculate the total NPV of the profit stream generated over the battery’s life, using the appropriate financial discount rate.
5. **Identify the Optimum:** Plot the resulting NPV against the candidate c_{base}^{aging} values. The value that corresponds to the peak of this curve is the optimal aging cost.[1]

This empirically determined optimal c^{aging} is the value that should be used in the operational MILP for real-world deployment. This rigorous, simulation-based tuning process ensures that the BESS’s day-to-day operational strategy is perfectly aligned with the overarching strategic goal of maximizing its long-term return on investment.

Table 5: Comprehensive Glossary for the Phase II MILP Formulation

	Category	Symbol	Definition	Unit
Sets & Indices				
t, T			Index and set of time intervals in the optimization horizon	-
j, J			Index and set of segments for piecewise linear functions	-
i, Ω^B			Index and set of network buses	-
bat, Ω^{bat}			Index and set of BESS units	-
Parameters				
E^{rate}			Nominal energy capacity of the BESS	MWh
$\overline{P^{ch}}, \overline{P^{dch}}$			Maximum charge and discharge power ratings	MW
η^{ch}, η^{dis}			Charge and discharge efficiencies	p.u.
$\overline{SoC}, \underline{SoC}$			Maximum and minimum State of Charge	p.u. or MWh
R			Battery cell replacement cost	\$/MWh
c^{aging}			Tunable aging cost hyperparameter	\$/ (unit LoH)
λ_t^e, λ_t^q			Forecasted prices for energy and reserve	\$/MWh, \$/MW
$P_{i,t}^{Load}, Q_{i,t}^{Load}$			Active and reactive load at bus i	MW, MVAR
R_{ij}, X_{ij}			Resistance and reactance of line ij	Ω
$\overline{V}, \underline{V}$			Maximum and minimum bus voltage limits	p.u. or V
I_{ij}			Thermal limit (maximum current) of line ij	A
Decision Variables				
$P_{bat,t}^{ch}, P_{bat,t}^{dch}$			Charge and discharge power of the BESS	MW
g_t, d_t			Generic discharge and charge power	MW
$SoC_{bat,t}$			State of Charge of the BESS	MWh
v_t			Binary variable for operating mode (1=discharge, 0=charge)	-
q_t			Reserve capacity provided by the BESS	MW
$LoH_t^{calendar}$			Calendar Loss of Health in time t	p.u.
LoH_t^{cyclic}			Cyclic Loss of Health in time t	p.u.
$\lambda_{t,i}^{cal}$			SOS2 variables for linearizing calendar aging	-
$P_{i,t}^G, Q_{i,t}^G$			Active/Reactive power from grid at bus i	MW, MVAR
$P_{ij,t}, Q_{ij,t}$			Active/Reactive power flow on line ij	MW, MVAR
$u_{i,t}$			Voltage squared at bus i	V^2

Conclusion

The proposed Phase II BESS operational model represents a significant leap forward from simpler approaches. By synthesizing state-of-the-art techniques from across the research landscape, it establishes a framework that is simultaneously physically realistic, economically rational, and robust to real-world forecast errors.

The key advancements can be summarized across three pillars:

1. **High-Fidelity Degradation Modeling:** The model moves beyond simplistic cycle counting to a unified formulation that captures both the SOC-dependent nature of calendar aging and the path-dependent, cumulative damage of cyclic aging through a linearized Rain-Flow Cycle Counting method. This ensures dispatch decisions are based on a comprehensive understanding of their long-term impact on asset health.
2. **Adaptive Optimization Architecture:** The adoption of a Deterministic Model Predictive Control (DMPC) framework provides a robust and computationally efficient method for real-world operation. The MPC outer loop provides an adaptive mechanism to correct for forecast errors and model mismatch by incorporating real-time information, ensuring operational viability.
3. **Strategic Lifetime Value Maximization:** The model reframes the economic objective from short-term profit maximization to long-term NPV maximization. It achieves this by treating the aging cost not as a fixed input, but as a tunable, strategic hyperparameter. This parameter is optimized via full-lifetime simulations to align the BESS’s operational aggressiveness with external market conditions and financial goals.

The optional integration of a linearized AC-OPF further extends the model’s capability, enabling the BESS to function as an active grid component capable of providing valuable local services. This modularity allows the framework to be tailored to diverse business cases, from large-scale wholesale market participation to localized, distribution-level grid support.

Ultimately, this framework provides a comprehensive blueprint for deploying and operating a BESS not merely as an energy storage device, but as a sophisticated financial asset. Its implementation will enable owners and operators to unlock the maximum possible economic value over the asset’s entire lifetime, navigating the complexities of modern electricity markets with a robust, intelligent, and forward-looking strategy.

10 Appendix: Knowledge Hub

10.1 The Asset: Mastering the the Huawei LUNA2000-4.5MWH-2H1 Smart String ESS

Fixed parameters:

- Nominal Capacity: 4,472 kWh
- Rated Power: 2,236 kW

10.2 Terminology

- Investment Terminology:
 - ROI (Return on Investment):
 - WACC (Weighted Average Cost of Capital):
 - Inflation Rate:
- Market Terminology:
 - Day-Ahead Wholesale Market: A blind auction that closes at 12:00 CET on the day before delivery (Day D-1). It determines the price for all 15-minute intervals of the following day (Day D).
 - Frequency Containment Reserve (FCR): A daily auction for capacity, not energy. The auction closes at 8 a.m. on Day $D - 1$ for six distinct 4-hour product blocks covering day D . Bids must be symmetric, meaning the BESS must offer an equal amount of capacity for both positive (discharging to counter low frequency) and negative (charging to counter high frequency) reserves. The minimum bid size is 1 MW.
 - Automatic Frequency Restoration Reserve (aFRR): The aFRR market is a two-stage process involving a capacity market and an energy market. The Balancing Capacity Market (BCM) auction closes at 9:00 on Day $D - 1$ for 4-hour product blocks. The Balancing Energy Market

(BEM) involves near-real-time auctions that close 25 minutes before each 15-minute delivery interval. Unlike FCR, aFRR bids are asymmetric, allowing the BESS to offer different amounts of capacity for positive (upward) and negative (downward) reserves. The minimum bid size is 1 MW.

10.3 What is BESS and EMS?

10.3.1 Technical Overview

At its core, a Battery Energy Storage System (BESS) is a large, rechargeable battery. Its main job on the power grid is to absorb electrical energy, store it, and then release it when it's most needed or most profitable.

If the BESS is the muscle, the Energy Management System (EMS) is the brain that tells the BESS what to do and when. It constantly analyzes data, such as:

- Market Electricity Prices
- Battery's State of Charge (SoC)
- Grid Frequency
- Physical Limits of the Battery (like C-rate)

A BESS **cannot simultaneously charge and discharge**. At any given moment, the net flow of power is either into the battery (charging) or out of the battery (discharging). While it can switch between these states very quickly (often in milliseconds), they are mutually exclusive actions.

Besides just buying and selling in the day-ahead market, your BESS can make money in another, more subtle way: by helping to keep the grid stable. These are called **ancillary services**, and they include things like Frequency Containment Reserve (FCR) and automatic Frequency Restoration Reserve (aFRR):

1. **Frequency Containment Reserve (FCR):** The grid has a "heartbeat" or frequency (50 Hz in Europe). FCR is the ultra-fast, automatic response to tiny deviations in this frequency. A BESS providing FCR is paid to be on standby, ready to inject or absorb tiny amounts of power instantly to stabilize the grid's heartbeat.
2. **Automatic Frequency Restoration Reserve (aFRR):** This is the second line of defense. If a bigger event happens (like a power plant suddenly going offline) and the frequency shifts more significantly, aFRR is activated to push it back to the target 50 Hz.

*FCR is like the **cruise control** in a car. It makes constant, tiny, automatic adjustments to the engine to keep your speed perfectly steady, without you having to think about it; aFRR is you, the driver, noticing you're slowing down on a hill and gently pressing the accelerator to get back to your desired speed. It's a slightly slower but more powerful and deliberate correction.*

These two parameters define the physical capabilities and warranty constraints of the BESS

1. **C-Rate:** The C-rate is a measure of how quickly a battery can be charged or discharged relative to its total capacity. A 1C rate means the battery can be fully charged or discharged in one hour. A 0.5C rate means it would take two hours, etc.
2. **(Daily) Cycle Limit:** This is a warranty constraint that limits how much you can use the battery each day to prevent it from degrading too quickly. One "cycle" is equivalent to one full charge and one full discharge. Your project mentions limits of 1.0, 1.5, and 2.0. A limit of 1.0 means you can use the battery's full capacity once per day. A limit of 2.0 means you could, for instance, fully charge and discharge it twice.

10.3.2 Economic Overview - ROI, WACC, & Inflation

These three metrics help you compare the project's long-term profitability across the five different European countries.

1. **Return on Investment (ROI):** This is the ultimate scorecard for your project. It's a simple ratio that tells you how much profit you made relative to the initial cost. *Formula:*

$$\text{ROI} = \frac{\text{Total Profit over 10 years}}{\text{Initial Investment Cost}} \times 100\%$$

Your goal is to find the country that offers the highest ROI over a 10-year period.

2. **Weighted-Average Cost of Capital (WACC):** This is the average interest rate a company has to pay to borrow the money needed to build the BESS. *Analogy:* Think of it as the project's "cost of money." A country with a high WACC is a more expensive place to finance the project. A higher WACC will eat into your profits and therefore lower your ROI.
3. **Inflation Rate:** This is the rate at which money loses its value over time. EUR 100 in your pocket today will buy less in a year. When you calculate your 10-year revenue, you have to account for inflation. The profits you make in Year 9 are worth less than the profits you make in Year 1. A country with high inflation will see its future revenues become less valuable more quickly.

Table 3: Model Sets, Parameters, and Decision Variables

Symbol	Definition	Unit	Type
Sets & Indices			
T	Set of 15-minute time intervals, $t \in T = \{1, \dots, 35040\}$	-	Set
B	Set of 4-hour ancillary service blocks, $b \in B$	-	Set
D	Set of 24-hour days, $d \in D$	-	Set
Parameters			
$P_{DA}(t)$	Day-ahead electricity price in interval t	EUR/MWh	Input
$P_{FCR}(b)$	FCR capacity price in block b	EUR/MW	Input
$P_{aFRR}^{pos}(b)$	Positive aFRR capacity price in block b	EUR/MW	Input
$P_{aFRR}^{neg}(b)$	Negative aFRR capacity price in block b	EUR/MW	Input
E_{nom}	Nominal energy capacity of the BESS	kWh	Input
P_{max}^{config}	Maximum charge/discharge power for the selected configuration	kW	Input
η_{ch}, η_{dis}	Charging and discharging efficiencies	-	Input
SOC_{min}, SOC_{max}	Min/max state of charge as a fraction of E_{nom}	-	Input
N_{cycles}	Daily cycle limit for the selected configuration	-	Input
e_{nom}^{init}	Initial state of charge at the start of the simulation period	kWh	Input
Δt	Duration of a time interval (0.25)	h	Constant
Δb	Duration of an ancillary service block (4)	h	Constant
$MinBid_{fcr}$	Minimum bid size for FCR market	MW	Input
$MinBid_{afr}$	Minimum bid size for aFRR market	MW	Input
Decision Variables			
$p_{ch}(t)$	Power used to charge the BESS in interval t	kW	Continuous ≥ 0
$p_{dis}(t)$	Power discharged from the BESS in interval t	kW	Continuous ≥ 0
$e_{soc}(t)$	Energy stored in the BESS at the	kWh	Continuous ≥ 0

WACC	Value		
Inflation Rate	value		
Discount Rate	Value		
Yearly Profits (2024)	Value		
Year		Initial Investment [kEUR/MWh]	Yearly profits [kEUR/MWh]
	2023		
	2024		
	2025		
	2026		
	2027		
	2028		
	2029		
	2030		
	2031		
	2032		
	2033		

Table 4: Comparative Analysis of Degradation Modeling Techniques

Feature	Xu et al. [1]	Collath et al. [1]	Soleimani et al. [1]	Proposed Phase II Model
Primary Aging Mechanism(s)	Cyclic Aging	Calendar & Cyclic Aging	Cyclic Aging	Calendar & Cyclic Aging
Key Stress Factors	Cycle Depth	SOC (Calendar), DOC & C-rate (Cyclic)	Cycle History (DoD)	SOC (Calendar), Cycle History (DoD)
Mathematical Method	Piecewise Linear Cost Function	Piecewise Linearization via SOS2	Piecewise Linearization of Cycle-to-Failure Curve	Combination of SOS2 (Calendar) and Piecewise Linearization of Cycle-to-Failure Curve (Cyclic)
Integration into Optimization	Indirect (Marginal Cost per MWh)	Direct (Capacity Loss per time step)	Direct (Loss of Health per cycle)	Direct (Loss of Health from both mechanisms)
Computational Complexity	Low (LP/MILP)	Medium (MILP with SOS2)	Medium (MILP)	Medium-High (MILP with SOS2)